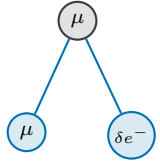
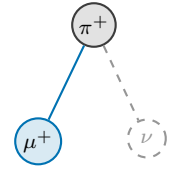
A. continuing particle



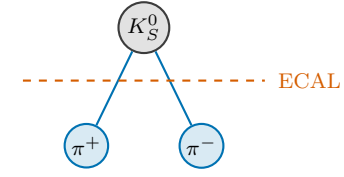
stored daughters carry $< 50\%$ of the energy (here 0.01%: ionisation in the muon system)
$\Rightarrow$ the μ **is terminal**, clustered with its full momentum; the δe^- is a constituent too

B. decay / conversion



daughters carry 100% of the energy
$\Rightarrow$ the π^+ is **replaced** by the μ^+; the ν is dropped (as in GenJetNoNu)
this is where the decay-in-flight response comes from

C. born inside the calorimeter



the pions never cross the tracker/calo boundary, so Geant4's `crossedBoundary` flag is not set, but the sim hits reference them
$\Rightarrow$ the terminal rule, not the flag, defines the particle lists (jets, ideal PF)

terminal $\Leftrightarrow$ no stored daughter, or $\sum_{\text{daughters}} E_d < 0.5\,E$ (`SimTrack_dauEnergyFrac` < 0.5)